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Inventor(s)	Lao; Chen-Hsiang et al.

Semiconductor package

Abstract

A semiconductor package includes a first heat dissipation plate, a second heat dissipation plate, a plurality of heat generating assemblies, and a plurality of fixture components. The first heat dissipation plate has a first upper surface and a first lower surface. The first heat dissipation plate includes first through holes extended from the first upper surface to the first lower surface. The second heat dissipation plate has a second upper surface and a second lower surface. The second heat dissipation plate includes second through holes extended from the second upper surface to the second lower surface. The heat generating assemblies are disposed between the first heat dissipation plate and the second heat dissipation plate. The fixture components include fix screws and nuts. The fix screws penetrate through the first heat dissipation plate and the second heat dissipation plate along the first through holes and the second through holes.

Inventors: Lao; Chen-Hsiang (New Taipei, TW), Chiu; Yuan-Sheng (Miaoli, TW), Li; Hung-Chi (Taipei, TW), Ku; Shih-Chang (Taipei, TW), Lin; Tsung-Shu (New Taipei, TW)

Applicant: Taiwan Semiconductor Manufacturing Company, Ltd. (Hsinchu, TW)

Family ID: 85286681

Assignee: Taiwan Semiconductor Manufacturing Company, Ltd. (Hsinchu, TW)

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Primary Examiner: Gauthier; Steven B

Assistant Examiner: Chan; Candice

Attorney, Agent or Firm: JCIPRNET

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a divisional application of and claims the priority benefit of a prior application Ser. No. 17/461,920, filed on Aug. 30, 2021 and now allowed. The entirety of each of the above-mentioned patent applications is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

(1) Semiconductor devices and integrated circuits used in a variety of electronic apparatus, such as cell phones and other mobile electronic equipment, are typically manufactured on a single semiconductor wafer. The dies of the wafer may be processed and packaged with other

semiconductor devices or dies at the wafer level, and various technologies have been developed for the wafer level packaging (e.g., formation of redistribution circuit structure/layer).

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.
- (2) FIG. 1 is a schematic view illustrating a semiconductor package in accordance with some embodiments of the disclosure.
- (3) FIG. 2 is a schematic three-dimensional (3D) view illustrating a first heat dissipation plate of the semiconductor package in FIG. 1 in accordance with some embodiments of the disclosure.
- (4) FIG. 3 is a schematic view illustrating a first heat dissipation plate of the semiconductor package 10 illustrated in FIG. 1 in accordance with some embodiments of the disclosure.
- (5) FIG. 4A is a schematic view illustrating a manufacturing process of a semiconductor package in accordance with some embodiments of the disclosure.
- (6) FIG. 4B is a schematic view illustrating a manufacturing process of a semiconductor package in accordance with some embodiments of the disclosure.
- (7) FIG. 4C is a schematic view illustrating a manufacturing process of a semiconductor package in accordance with some embodiments of the disclosure.
- (8) FIG. 4D is a schematic view illustrating a manufacturing process of a semiconductor package in accordance with some embodiments of the disclosure.
- (9) FIG. 4E is a schematic view illustrating a manufacturing process of a semiconductor package in accordance with some embodiments of the disclosure.
- (10) FIG. 5 is a schematic view illustrating a semiconductor package in accordance with some embodiments of the disclosure.
- (11) FIG. 6A is a schematic view illustrating a manufacturing process of a semiconductor package in accordance with some embodiments of the disclosure.
- (12) FIG. 6B is a schematic view illustrating a manufacturing process of a semiconductor package in accordance with some embodiments of the disclosure.
- (13) FIG. 6C is a schematic view illustrating a manufacturing process of a semiconductor package in accordance with some embodiments of the disclosure.
- (14) FIG. 6D is a schematic view illustrating a manufacturing process of a semiconductor package in accordance with some embodiments of the disclosure.
- (15) FIG. 6E is a schematic view illustrating a manufacturing process of a semiconductor package in accordance with some embodiments of the disclosure.
- (16) FIG. 6F is a schematic view illustrating a manufacturing process of a semiconductor package in accordance with some embodiments of the disclosure.

DETAILED DESCRIPTION

- (17) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components, values, operations, materials, arrangements, or the like, are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. Other components, values, operations, materials, arrangements, or the like, are contemplated. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and

second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(18) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(19) In addition, terms, such as “first,” “second,” “third,” “fourth,” and the like, may be used herein for ease of description to describe similar or different element(s) or feature(s) as illustrated in the figures, and may be used interchangeably depending on the order of the presence or the contexts of the description.

(20) Other features and processes may also be included. For example, testing structures may be included to aid in the verification testing of the 3D packaging or 3DIC devices. The testing structures may include, for example, test pads formed in a redistribution layer or on a substrate that allows the testing of the 3D packaging or 3DIC, the use of probes and/or probe cards, and the like. The verification testing may be performed on intermediate structures as well as the final structure. Additionally, the structures and methods disclosed herein may be used in conjunction with testing methodologies that incorporate intermediate verification of known good dies to increase the yield and decrease costs.

(21) FIG. 1 is a schematic view illustrating a semiconductor package **10** in accordance with some embodiments of the disclosure. In some embodiments, the semiconductor package **10** includes a first heat dissipation plate **110**, a second heat dissipation plate **120**, a plurality of heat generating assemblies **200**, and a plurality of fixture components **144**, **146**.

(22) In some embodiments, as shown in FIG. 1, the first heat dissipation plate **110** has a first upper surface **111** and a first lower surface **113**. Moreover, the first heat dissipation plate **110** includes first through holes **114**, **116**, and each of the first through holes **114**, **116** is extended from the first upper surface **111** to the first lower surface **113**.

(23) In the present embodiment, the second heat dissipation plate **120** has a second upper surface **121** and a second lower surface **123**. As shown in FIG. 1, the second heat dissipation plate **120** includes second through holes **124**, **126**. In some embodiments, each of the second through holes **124**, **126** is extended from the second upper surface **121** to the second lower surface **123** and respectively aligned with the first through holes **114**, **116**.

(24) In some embodiments, the heat generating assemblies **200** are respectively disposed between the first heat dissipation plate **110** and the second heat dissipation plate **120**. Referring to FIG. 1, each of the heat generating assemblies **200** includes a first heat generating component **210** and a second heat generating component **220**. In some embodiments, the first heat generating component **210** is, for example, a voltage regulator module (VRM), and the second heat generating component **220** includes, for example, a semiconductor die. In some embodiments, the VRM **210** is disposed on the first upper surface **111** of the first heat generating plate **110**, and the semiconductor die **220** is disposed on the second lower surface **123** of the second heat generating plate **120**.

(25) Referring to FIG. 1, in some embodiments, the fixture components **144**, **146** respectively include fix screws **144a**, **146a** and nuts **144b**, **146b**. As shown in FIG. 1, the fix screws **144a**, **146a** respectively include screw heads **144a1**, **146a1** and tread portions **144a2**, **146a2** connected with each other. The fix screws **144a**, **146a** respectively penetrate through the first heat dissipation plate **110** and the second heat dissipation plate **120** along the first through holes **114**, **116** and the second through holes **124**, **126**. In addition, the thread portions **144a2**, **146a2** are respectively threaded

with the nuts **144b**, **146b** and protruded therefrom along a direction away from the first heat dissipation plate **110**.

(26) In some embodiments, referring to FIG. 1, the screw heads **144a1**, **146a1** of the fix screws **144**, **146** are pressed against the second upper surface **121** of the second heat dissipation plate, and the nuts **144b**, **146b** are fastened with the thread portions **144a2**, **146a2** through applying various torques to fixedly secure the heating generating assemblies **200** between the first heat dissipation plate **110** and the second heat dissipation plate **120**.

(27) In some embodiments, the semiconductor package **10** further includes a back plate **170**. The back plate **170** is disposed below the first lower surface **113** of the first heat dissipation plate **110**. Referring to FIG. 1, the first heat dissipation plate **110** is disposed between the VRMs **210** of the heat generating assemblies **200** and the back plate **170**. As shown in FIG. 1, the tread portions **144a2**, **146a2** of the fix screws **144**, **146** respectively penetrate through the back plate **170**, and the tread portions **144a2**, **146a2** are protruded from the bottom of the back plate **170**. In some embodiments, as illustrated in FIG. 1, the back plate **170** includes through holes **174**, **176**, and the tread portions **144a2**, **146a2** are inserted through the through holes **174**, **176** and protruded therefrom.

(28) In some embodiments, the nuts **144b**, **146b** are respectively treaded with the thread portions **144a2**, **146a2** in different strokes and pressed against the bottom of the back plate **170** to securely fasten the back plate **170** to the first heat dissipation plate **110** and the heat generating assemblies **200**. Referring to FIG. 1, the back plate **170** further includes through vias **172** disposed between the through holes **174**, **176**. In the present embodiments, the semiconductor package **10** also includes a plurality of stop screws **160** respectively accommodate in the through vias **172**. In some embodiments, the stop screws **160** are respectively pressed against the first lower surface **113** of the first heat dissipation plate **110** to upwardly push the first heat dissipation plate **110** to the deformed positions for uniformly contacting the bottoms of the VRMs **210**. In some embodiments, the back plate **170** can be formed by for example a stainless-steel material having greater hardness to avoid being deformed by the press forces applied from the fixture components **144**, **146**.

(29) In some embodiments, referring to FIG. 1, the first heat dissipation plate **110** further includes first lug bosses **115** formed on the first upper surface **111**, and the VRMs **210** are respectively disposed on the first lug bosses **115**. In some embodiments, a first thermal interface material (TIM) layer **132** is disposed between the VRMs **210** and the first lug bosses **115** to enhance thermal dissipation. In some embodiments, a vertical height of the of the first lug bosses **115** from the first upper surface **111** is in a range from about 1 mm to about 3 mm.

(30) In some embodiments, as shown in FIG. 1, the second heat dissipation plate **120** can include second lug bosses **125** formed on the second lower surface **123**, and the semiconductor dies **220** are respectively disposed on the second lug bosses **125**. In some embodiments, a second thermal interface material (TIM) layer **134** is disposed between the semiconductor dies **220** and the second lug bosses **125**. In some embodiments, a vertical height of the of the second lug bosses **125** from the second lower surface **123** is in a range from about 1 mm to about 3 mm.

(31) As shown in FIG. 1, the semiconductor dies **220** are respectively encapsulated by insulating encapsulation material **215**. In some embodiments, the insulating encapsulation material **215** is a molding compound formed by a molding process. In some embodiments, the insulating encapsulation material **215**, for example, may include polymers (such as epoxy resin, phenolic resins, silicon-containing resins, or other suitable resins), and dielectric materials having low permittivity (Dk) and low loss tangent (Df) properties, or other suitable materials. In some embodiments, a wafer substrate **230** is disposed between the VRMs **210** and the semiconductor dies **220**. In some embodiments, a plurality of connecting elements **225** can be further disposed between the VRMs **210** and the wafer substrate **230** to form conductive connections therebetween. In some embodiments, the connecting elements **225** are also encapsulated by the insulating encapsulation material **215**.

(32) In the present embodiments, the nuts **144b**, **146b** are treaded with the tread portions **144a2**, **146a2** of the fix screws **144a**, **146a** by various applied torques, so that the screw heads **144a1**, **146a1** and the nuts **144b**, **146b** are respectively pressed against the first lower surface **113** and the second upper surface **121** to uniformly attach the first lug bosses **115** and the second lug bosses **125** respectively to the bottom of VRMs **210** and the upper surfaces of the semiconductor dies **220** through the first TIM layer **132** and the second TIM layer **134**.

(33) In some embodiments, the first TIM layer **132** and the second TIM layer **134** respectively include thermal conductive polymer materials for enhancing thermal conduction or thermal dissipation across the interfaces between the bottoms of the VRMs **210** and the first lug bosses **115** of the first heat dissipation plate **110** and between the upper surfaces of the semiconductor dies **220** and the second lug bosses **125** of the second heat dissipation plate **120**. Hence, the VRMs **210** and the semiconductor dies **220** are pressed on and in contact with the first heat dissipation plate **110** and the second heat dissipation plate **120** through the interfaces therebetween in a uniform manner. In the present embodiments, the press forces applied on the first TIM layer **132** and the second TIM layer **134** can be uniformly spread throughout the entire interfaces with the first lug bosses **115** and the second lug bosses **125** instead of being concentrate on the central areas of the interfaces. Through the above configurations, efficiency and uniformity of thermal conduction and dissipation of the heat generated from the heat generating assemblies **200** including the VRMs **210** and the semiconductor dies **220** can be further enhanced.

(34) FIG. 2 is a schematic three-dimensional (3D) view illustrating a first heat dissipation plate **110** of the semiconductor package **10** in FIG. 1 in accordance with some embodiments of the disclosure. Referring to FIG. 1 and FIG. 2, in some embodiments, a diameter size of the through holes **114** located around the peripheral area of the first heat dissipation plate **110** is greater than a diameter size of the through holes **116** located in the central area of the first heat dissipation plate **110**. In some embodiments, referring to FIG. 1 and FIG. 2, the through holes **114** disposed in peripheral area of the first heat dissipation plate **110** can accommodate the fix screws **144** having a diameter size of 4 mm substantially. Moreover, the through holes **115** disposed in central area of the first heat dissipation plate **110** can accommodate the fix screws **146** having a diameter size of 3 mm substantially.

(35) Referring to FIG. 1 and FIG. 2, the first through holes **114** of the first heat dissipation plate **110**, the second through holes **124** of the second heat dissipation plate **120**, and the through holes **174** of the back plate **170** are vertically aligned with each other correspondingly and have the same diameter size to accommodate the fix screws **144** inserted therein. In addition, the first through holes **116** of the first heat dissipation plate **110**, the second through holes **126** of the second heat dissipation plate **120**, and the through holes **176** of the back plate **170** are vertically aligned with each other correspondingly and have the same diameter size to accommodate the fix screws **146** inserted therein.

(36) As shown in FIG. 2, the first heat dissipation plate **110** further includes a plurality of blind vias **119** disposed on the first lower surface **113**, which accommodate the stop screws **160** extruded from the back plate **170** and respectively press against the first heat dissipation plate **110**. In some embodiments, the first heat dissipation plate **110** can be formed by a flexible metal material (e.g., 3003 aluminum alloy or O temper 6000 series aluminum alloy) having a Young's modulus about 50% lower than the commonly used aluminum alloy comprising Cu or Mg, for example, 6061 aluminum alloy. Hence, the first heat dissipation plate **110** is adapted to be pressed and deformed by the stop screws **160** and the fixture components **144**, **146**.

(37) FIG. 3 is a schematic view illustrating a first heat dissipation plate **110** of the semiconductor package **10** illustrated in FIG. 1 in accordance with some embodiments of the disclosure. In some embodiments, the first heat dissipation plate **110** can be a cold plate having a cooling fluid (not illustrated) circulated therein. Referring to FIG. 3, in some embodiments, the first heat dissipation plate **110** can include a plurality of cooling fluid pipes **117** and cooling fluid tanks **118**. Referring to

FIG. 2 and FIG. 3, in some embodiments, the cooling fluid enters the cooling fluid pipes **117** and the cooling fluid tanks **118** through an inlet **112a** and exits through an outlet **112b**. As shown in FIG. 3, the cooling fluid pipes **117** are curvedly extended in the first heat dissipation plate **110** to avoid interference with the pins of the VRMs **210** protruded from the bottom of the first heat dissipation plate **110**. In some embodiments, the heat generated by the VRMs **210** can be transmitted and dissipated into the cooling fluid circulated in the first heat dissipation plate **110** and carried away by the cooling fluid.

(38) FIG. 4A is a schematic view illustrating a manufacturing process of a semiconductor package **10** illustrated in FIG. 1 in accordance with some embodiments of the disclosure. In some embodiments, in a manufacturing process of the semiconductor package **10**, a plurality of heat generating assemblies **200** respectively including the VRMs **210** and the semiconductor dies **220** are provided. Moreover, the wafer substrate **230** including through holes **234**, **236** is disposed between the VRMs **210** and the semiconductor dies **220**. In the present embodiments, the wafer substrate **230** with the integrated fan-out (inFO) structure is provided, and the VRMs **210** can be soldered joint on the wafer substrate **230** through the connecting elements **225**.

(39) FIG. 4B is a schematic view illustrating a manufacturing process of a semiconductor package **10** in accordance with some embodiments of the disclosure. Referring to FIG. 4B, the first heat dissipation plate **110** with the back plate **170** disposed below the first lower surface **113** thereof is provided. In some embodiments, the first heat dissipation plate **110** includes the first lug bosses **115** protruded from the first upper surface **111** of the first heat dissipation plate **110**. In addition, the first TIM layer **132** is formed on the first lug bosses **115**. As shown in FIG. 4B, the first heat dissipation plate **110** includes the through holes **114**, **116** formed therein, and the back plate **170** includes the through holes **174**, **146** formed therein respectively aligned with the through holes **114**, **116**. In addition, the through vias **172** are formed in the bottom of the back plate **170** to accommodate the stop screws **160**. In some embodiments, the strokes of each of the stop screws **160** can be respectively adjusted to contact and press against the first lower surface **113** of the first heat dissipation plate **110**.

(40) FIG. 4C is a schematic view illustrating a manufacturing process of a semiconductor package **10** in accordance with some embodiments of the disclosure. In some embodiments, referring to FIG. 4A to FIG. 4C, the heat generating assemblies **200** are respectively attached to the first lug bosses **115** of the first heat dissipation plate **110** with the first TIM layer **132** disposed therebetween. Depending on the heights of the VRMs **210**, gaps might be still existed between some of the VRMs **210** and the first TIM layer **132** as the heat generating assemblies **200** are attached to the first TIM layer **132** and the first lug bosses **115** at the beginning of the process.

(41) FIG. 4D is a schematic view illustrating a manufacturing process of a semiconductor package **10** in accordance with some embodiments of the disclosure. As shown in FIG. 4D, the fix screws **144a**, **146a** are inserted into the through holes **114**, **116** of the first heat dissipation plate **110**, the through holes **124**, **126** of the second heat dissipation plate **120**, the through holes **234**, **236** of the wafer substrate **230**, and the through holes **174**, **176** of the back plate **170**. Moreover, the tread portions **144a2**, **146a2** of the fix screws **144a**, **146a** are respectively threaded and pre-fixed through the nuts **144b**, **146b**. As the step of manufacturing process shown in FIG. 4D, the fix screws **144a**, **146a** are still not fully tightened with the nuts **144b**, **146b**, and thus the gaps GP1, GP2 having different heights are existed between the VRMs **210** of the heat generating assemblies **200** and the first TIM layer **132** formed on the first lug bosses **115**.

(42) FIG. 4E is a schematic view illustrating a manufacturing process of a semiconductor package **10** in accordance with some embodiments of the disclosure. As shown in FIG. 4E, the thread portions **144a2**, **146a2** of the fix screws **144a**, **146a** threaded in the nuts **144b**, **146b** are respectively further fastened by applying different torques, and the nuts **144b**, **146b** are respectively pressed against the bottom of the back plate **170**. In addition, the screw heads **144a1**, **146a1** are respectively pressed against the second upper surface **121** of the second heat dissipation plate **120**.

to fasten the fix screws **144a**, **146a**.

(43) Referring again to FIG. **4E**, through further fastening the fix screws **144a**, **146a** to the nuts **144b**, **146b** respectively by different torques and strokes to compensate the height differences between the various VRMs **210**, the gaps GP1, GP2 respectively existed between the VRMs **210** and the first TIM layer **132** on the first lug bosses **115** can be eliminated, so that the VRMs **210** can be uniformly attached to and pressed on the first TIM layer **132** and the first lug bosses **115** of the first heat dissipation plate **110**.

(44) Referring again to FIG. **4D** and FIG. **4E**, in the present embodiments, the VRMs **210** respectively have different heights along a stacking direction of the first heat dissipation plate **110** and the second heat dissipation plate **120**. The difference of the heights of the VRMs **210** are originated from the original production tolerance thereof. When the fix screws **144a**, **146a** are further fastened with the nuts **144b**, **146b** by applying different strokes and torques, the nuts **144b**, **146b** are pressed against the first lower surface **113** of the first heat dissipation plate **110**. In addition, the production tolerance of the VRMs **210** can be further compensated through disposing the first TIM layer **132** having various thicknesses on the first lug bosses **115**. As a result, the first heat dissipation plate **110** can be pushed upward by the nuts **144b**, **146b** and the stop screws **160** to uniformly attach the first lug bosses **115** to the bottom of the VRMs **210**. Hence, the heat generated by the VRMs **210** can be uniformly transmitted and dissipated to the first heat dissipation plate **110** through the first TIM layer **132** disposed therebetween.

(45) FIG. **5** is a schematic view illustrating a semiconductor package **20** in accordance with some embodiments of the disclosure. Referring to FIG. **5**, in the current embodiments, the difference between the semiconductor package **20** and the semiconductor package **10** is that the semiconductor package **20** includes a back plate **180**, and the back plates **180** includes a plurality of blind holes **184**, **186** for accommodating fixture components **154**, **156**. Referring to FIG. **5**, in some embodiments, the fixture components **154**, **156** respectively include fix screws **154a**, **156a** and nuts **154b**, **156b**.

(46) In the present embodiments, the fix screws **154a**, **156a** includes an upper tread portions **154a2**, **156a2** and lower tread portions **154a1**, **156a1**. Referring again to FIG. **5**, the lower tread portions **154a1**, **156a1** respectively treaded into the blind holes **184**, **186** of the back plate **180**, and the upper tread portion **154a2**, **156a2** respectively treaded in the nuts **154b**, **156b**. In the present embodiments, the nuts **154b**, **156b** are respectively pressed against the second upper surface **121** of the second heat dissipation plate **120**. In the present embodiments, the lower tread portions **154a1**, **156a1** treaded into the blinded holes **184**, **186** of the back plate **180** to secure the fix screws **154a**, **156a** in the back plate **180**.

(47) As shown in FIG. **5**, the upper tread portion **154a2**, **156a2** are treaded through and protruded from the nuts **154b**, **156b**. The nuts **154b**, **156b** pressed against the second upper surface **121** of the second heat dissipation plate **120** to push the second lug bosses **125** of the second heat dissipation plate **120** to uniformly attach on the second heat generating components **220**, for example, semiconductor dies.

(48) FIG. **6A** is a schematic view illustrating a manufacturing process of a semiconductor package **20** in accordance with some embodiments of the disclosure. In the manufacturing process of the semiconductor package **20**, a plurality of heat generating assemblies **200** respectively including the VRMs **210** and the semiconductor dies **220** are provided. Moreover, the wafer substrate **230** including through holes **234**, **236** can be disposed between the VRMs **210** and the semiconductor dies **220**. In the manufacturing process of the semiconductor package **20**, the wafer **230** with integrated fan-out (inFO) structure is provided, and the VRMs **210** can be soldered on the wafer **230**.

(49) FIG. **6B** is a schematic view illustrating a manufacturing process of a semiconductor package **20** in accordance with some embodiments of the disclosure. Referring to FIG. **6B**, the first heat dissipation plate **110** with the back plate **180** disposed below the first lower surface **113** thereof is

provided. In some embodiments, the first heat dissipation plate **110** includes the first lug bosses **115** protruded from the first upper surface **111** of the first heat dissipation plate **110**. In addition, the first TIM layer **132** is disposed on the first lug bosses **115**. As shown in FIG. **6B**, the first heat dissipation plate **110** includes the through holes **114**, **116**, and the back plate **180** includes the through holes **184**, **186** respectively aligned with the through holes **114**, **116** of the first heat dissipation plate **110**. The through vias **182** are further formed in the back plate **180** for accommodating the stop screws **160**. As shown in FIG. **6B**, the stroke of each of the stop screws **160** can be adjusted to contact and press against the first lower surface **113** of the first heat dissipation plate **110**.

(50) Referring again to FIG. **6B**, the blind holes **184**, **186** are respectively formed in the back plate **180** for accommodating the fix screws **152**, **154** protruded from the first lower surface **113** of the first heat dissipation plate **110**.

(51) FIG. **6C** is a schematic view illustrating a manufacturing process of a semiconductor package **20** in accordance with some embodiments of the disclosure. In some embodiments, referring to FIG. **6C**, a thread locking fluid **185**, for example an anaerobic adhesive, is filled into the blind holes **184**, **186** for firmly securing the fix screws **154**, **156** inside of the blind holes **184**, **186**.

(52) FIG. **6D** is a schematic view illustrating a manufacturing process of a semiconductor package **20** in accordance with some embodiments of the disclosure. Referring to FIG. **6D**, the heat generating assemblies **200** are respectively attached to the first lug bosses **115** of the first heat dissipation plate **110** with the first TIM layer **132** disposed therebetween. Depending on the heights of the VRMs **210**, gaps might be still existed between some of the VRMs **210** and the first TIM layer **132** as the heat generating assemblies **200** are attached to the first TIM layer **132** and the first lug bosses **115** at the beginning of the process.

(53) FIG. **6E** is a schematic view illustrating a manufacturing process of a semiconductor package **20** in accordance with some embodiments of the disclosure. As shown in FIG. **6E**, the fix screws **154**, **156** are respectively inserted through the through holes **114**, **116** of the first heat dissipation plate **110** and the through holes **124**, **126** of the second heat dissipation plate **120**. Moreover, the lower threaded portion **154a1**, **156a1** of the fix screws **154a**, **156a** are respectively threaded and pre-fixed into the blind holes **184**, **186**. In some embodiments, the nuts **154b**, **156b** are respectively treaded with the upper treaded portions **154a2**, **156a2** of the fix screws **154a**, **156a**, and the upper treaded portions **154a2**, **156a2** are further protruded from the nuts **154a**, **156b** along a direction away from the second heat dissipation plate **120**. As shown in FIG. **6E**, the fix screws **154**, **156** are not fully tightened in the nuts **154b**, **156b**, and thus a plurality of gaps GP3, GP4 having different heights may be existed between the various VRMs **210** of the heat generating assemblies **200** and the first TIM layer **132** on the first lug bosses **115** due to variation in the heights of the VRMs **210** originated from production tolerance.

(54) FIG. **6F** is a schematic view illustrating a manufacturing process of a semiconductor package **20** in accordance with some embodiments of the disclosure. As shown in FIG. **6F**, the fix screws **154a**, **156a** inserted through the nuts **154b**, **156b** are further treaded in various strokes and fastened by different torques. The nuts **154b**, **156b** are respectively pressed against the second upper surface **121** of the second heat dissipation plate **120**. In addition, the lower tread portions **154a1**, **156a1** are inserted into the blind holes **184**, **186** of the back plate **180** and firmly secured therein through the thread locking fluids **185**.

(55) Referring again to FIG. **6F**, through further fastening the fix screws **154a**, **156a** to the nuts **154b**, **156b**, the gaps GP3, GP4 respectively existed between the VRMs **210** and the first TIM layer **115** can be eliminated by applying various torques to the nuts **152b**, **154b**, so that the nuts **152b**, **154b** respectively different strokes in relative to the upper tread portions **154a2**, **156a2**. As a result, the VRMs **210** can be uniformly attached to and pressed against the first lug bosses **115** with the first TIM layer **132** disposed therebetween, and the upper surfaces of the semiconductor dies **220** are attached to the second lug bosses **125** of the second heat dissipation plate **120**.

(56) Referring again to FIG. 6E and FIG. 6F, in the present embodiments, when the fix screws **154a**, **156a** are further fastened with the nuts **154b**, **156b** by applying different strokes and torques through for examples a screw tightening machine, the nuts **154b**, **156b** are pressed against the second upper surface **121** of the second heat dissipation plate **120**. As a result, the second heat dissipation plate **120** can be pushed by the nuts **154b**, **156b** toward the heat generating assemblies **200**. The stop screws **160** are pressed on the first lower surface **113** of the first heat dissipation plate **110** to the deformed positions DP thereof. Through the above configurations, the bottom surfaces of the VRMs **210** are unfirmly attached to the first lug bosses **115** and first TIM layer **132** disposed therebetween, and the upper surfaces of the semiconductor dies **220** are uniformly attached to the second lug bosses **125**. Hence, the heat generated by the VRMs **210** and the semiconductor dies **220** can be uniformly transmitted and dissipated to the first heat dissipation plate **110** and the second heat dissipation plate **120** respectively through the first TIM layer **132** and the second TIM layer **134**. Through above steps of the manufacturing process, the semiconductor package **20** is fully formed.

(57) In accordance with some embodiments, a semiconductor package includes a first heat dissipation plate, a second heat dissipation plate, a plurality of heat generating assemblies, and a plurality of fixture components. The first heat dissipation plate has a first upper surface and a first lower surface. The first heat dissipation plate includes first through holes, and each of the first through holes is extended from the first upper surface to the first lower surface. The second heat dissipation plate has a second upper surface and a second lower surface. The second heat dissipation plate includes second through holes, and each of the second through holes is extended from the second upper surface to the second lower surface and aligned with one of the first through holes. The heat generating assemblies are disposed between the first heat dissipation plate and the second heat dissipation plate. Each of the heat generating assemblies includes a semiconductor die disposed on the second lower surface. The fixture components include fix screws and nuts. The fix screws respectively include screw heads and thread portions connected with each other, and the fix screws penetrate through the first heat dissipation plate and the second heat dissipation plate along the first through holes and the second through holes. The thread portions of the fix screws are respectively threaded through the nuts and protruded therefrom along a direction away from the first lower surface, and the screw heads are pressed against the second upper surface to fixedly secure the heat generating assemblies between the first heat dissipation plate and the second heat dissipation plate.

(58) In accordance with some embodiments, a semiconductor package includes a first heat dissipation plate, a plurality of first heat generating components, a second heat dissipation plate, a plurality of second heat generating components, a back plate, and a plurality of fixture components. The first heat dissipation plate has a first upper surface and a first lower surface. The first heat dissipation plate includes first lug bosses and first through holes. Each of the first through holes is extended from the first upper surface to the first lower surface, and each of the first lug bosses is protruded from the first upper surface. The first heat generating components are respectively disposed on the first lug bosses. The second heat dissipation plate has a second upper surface and a second lower surface. The second heat dissipation plate includes second lug bosses and second through holes. Each of the second through holes is extended from the second upper surface to the second lower surface, and each of the second lug bosses is protruded from the second lower surface. The second heat generating components are respectively disposed on the second lug bosses and between the first generating components and the second heat dissipation plate. The back plate is disposed below the first lower surface of the first heat generating components. The fixture components include fix screws and nuts. The fix screws penetrate through the first heat dissipation plate, the second heat dissipation plate, and the back plate along the first through holes and the second through holes, and the fix screws are protruded from the back plate along a direction away from the first lower surface. The fix screws are respectively threaded through the nuts, and the nuts

are pressed against the back plate to fixedly secure the first heat generating components and the second heat generating components between the first heat dissipation plate and the second heat dissipation plate.

(59) In accordance with some embodiments, a semiconductor package includes a first heat dissipation plate, a second heat dissipation plate, a plurality of heat generating assemblies, a back plate, and a plurality of fixture components. The first heat dissipation plate has a first upper surface and a first lower surface. The first heat dissipation plate includes first lug bosses and first through holes. Each of the first through holes is extended from the first upper surface to the first lower surface, and each of the first lug bosses is protruded from the first upper surface. The second heat dissipation plate has a second upper surface and a second lower surface. The second heat dissipation plate includes second lug bosses and second through holes. Each of the second through holes is extended from the second upper surface to the second lower surface, and each of the second lug bosses is protruded from the second lower surface. The heat generating assemblies are respectively disposed between the first lug bosses and the second lug bosses. The back plate is disposed below the first lower surface of the first heat dissipation plate. The back plate includes a plurality of blind holes respectively aligned with the first through holes and the second through holes. The fixture components include fix screws and nuts. The fix screws penetrate through the first heat dissipation plate and the second heat dissipation plate along the first through holes and the second through holes and treaded into the blind holes of the back plate. The fix screws are respectively threaded through the nuts to fixedly secure the heat generating assemblies between the first heat dissipation plate and the second heat dissipation plate.

(60) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A semiconductor package, comprising: a first heat dissipation plate having a first upper surface and a first lower surface, wherein the first heat dissipation plate comprises first lug bosses and first through holes, wherein each of the first through holes is extended from the first upper surface to the first lower surface, and each of the first lug bosses is protruded from the first upper surface; a plurality of first heat generating components, respectively disposed on the first lug bosses; a second heat dissipation plate having a second upper surface and a second lower surface, wherein the second heat dissipation plate comprises second lug bosses and second through holes, wherein each of the second through holes is extended from the second upper surface to the second lower surface, and each of the second lug bosses is protruded from the second lower surface; a plurality of second heat generating components, respectively disposed on the second lug bosses and between the first heat generating components and the second heat dissipation plate; a back plate disposed below the first lower surface of the first heat dissipation plate; and a plurality of fixture components, comprising fix screws and nuts, wherein the fix screws penetrate through the first heat dissipation plate, the second heat dissipation plate, and the back plate along the first through holes and the second through holes, and the fix screws are protruded from the back plate along a direction away from the first lower surface, wherein the fix screws are respectively threaded through the nuts, and the nuts are pressed against the back plate to fixedly secure the first heat generating components and the second heat generating components between the first heat dissipation plate and the second

heat dissipation plate.

2. The semiconductor package as claimed in claim 1, wherein each of the first heat generating components comprises a voltage regulator module.

3. The semiconductor package as claimed in claim 2, wherein the voltage regulator module of each of the first heat generating components has a different height along a stacking direction of the first heat dissipation plate and the second heat dissipation plate.

4. The semiconductor package as claimed in claim 1, wherein each of the second heat generating components comprises a semiconductor die.

5. The semiconductor package as claimed in claim 1, wherein a first thermal interface material layer is disposed between the first heat generating components and the first lug bosses.

6. The semiconductor package as claimed in claim 1, wherein gap heights between the first heat generating components and the first lug bosses are respectively different from each other.

7. The semiconductor package as claimed in claim 1, further comprising a wafer substrate disposed between the first heat generating components and the second heat generating components.

8. The semiconductor package as claimed in claim 1, wherein the second heat dissipation plate is a continuous plate extended in a lateral direction of the semiconductor package and fully covers the plurality of heat generating assemblies.

9. A semiconductor package, comprising: a first heat dissipation plate having a first upper surface and a first lower surface, wherein the first heat dissipation plate comprises first lug bosses and first through holes, wherein each of the first through holes is extended from the first upper surface to the first lower surface, and each of the first lug bosses is protruded from the first upper surface; a plurality of first heat generating components, respectively disposed on the first lug bosses; a second heat dissipation plate having a second upper surface and a second lower surface, wherein the second heat dissipation plate comprises second lug bosses and second through holes, wherein each of the second through holes is extended from the second upper surface to the second lower surface, and each of the second lug bosses is protruded from the second lower surface; a plurality of second heat generating components, respectively disposed on the second lug bosses and between the first heat generating components and the second heat dissipation plate; a back plate disposed below the first lower surface of the first heat dissipation plate; and a plurality of fixture components, comprising fix screws and nuts, wherein the fix screws penetrate through the first heat dissipation plate, the second heat dissipation plate, and the back plate along the first through holes and the second through holes, and the fix screws are protruded from the back plate along a direction away from the first lower surface, wherein the fix screws are respectively threaded through the nuts, and the nuts are pressed against the back plate to fixedly secure the first heat generating components and the second heat generating components between the first heat dissipation plate and the second heat dissipation plate, wherein each of the heat generating assemblies further comprises a voltage regulator module disposed on the first upper surface and between the semiconductor die and the first heat dissipation plate, wherein the voltage regulator modules respectively have different heights along a stacking direction of the first heat dissipation plate and the second heat dissipation plate.

10. The semiconductor package as claimed in claim 9, wherein the second heat dissipation plate is a continuous plate extended in a lateral direction of the semiconductor package and fully covers the plurality of heat generating assemblies.

11. The semiconductor package as claimed in claim 9, further comprising a wafer substrate disposed between the first heat generating components and the second heat generating components.

12. The semiconductor package as claimed in claim 9, wherein gap heights between the first heat generating components and the first lug bosses are respectively different from each other.
